2. LITERATURE REVIEW

2.1 Introduction

Software project success is generally measured using the triple constraint ('iron golden' triangle), namely cost goals, time goals and project performance in terms of project scope completion [5]. Therefore, an organization would want to have a software project scope finished within these constrains. Software companies are in constant search for tools and techniques that would assist project managers manage the triple constraint. Project scope management is one of the knowledge areas in which a project manager should have adequate skills competencies. Moreover, project scope management ensures that project stakeholders have the same understanding of the type of products the project will produce [2]. This is achieved through the use of scope definition and scope verification processes which are part of project scope management. The next section discusses the scope verification process.

2.2 Project Scope Verification

In broad terms, verification can be regarded as a 'process of determining whether a software development phase has been correctly carried out' [6]. On the other hand [3] defines verification as a process used to formalize project scope acceptance. Scope verification is a process that is carried out under specification analysis phase. In this phase client's requirements are analyzed and interpreted in order to produce a specification document which will contain project scope of work. Project scope verification is an important process in ensuring that the project team delivers exactly what the customer requested [5] and also in ensuring that project scope changes are minimal [2]. It is the process that formalizes the acceptance of the project scope. On the other hand, [3] regards verification as more than just a process which ensures deliverables conform to user requirements, but also as a quality assurance process too – ensuring that project work is complete and correct.

Various tools and techniques have been suggested for project scope verification process [2], [6] and these include:

- Inspection – the customer or user inspects the work after it is delivered.
- Prototyping – working replica of the planned system.
- Use case modeling – a tool used to model business events in order to gain better understanding of user requirements.
- Joint application design – a technique used to promote greater involvement of key project stakeholders in system development.
- Walkthrough – A document is carefully checked by a team of software professionals.

Any of these tools and techniques may be applied in the process of project scope verification. In order for project scope verification to proceed smoothly, scope definition has to be done thoroughly. A well defined project scope is a pre-cursor for successful project scope control, project success and customer satisfaction. The next section discusses work breakdown structure which plays an important role in defining a project scope.